Syracuse University
School of Information Studies

Week 5
Importing Data

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Flat Files

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Remote Data Access

Transactional Data Access

- Sometimes it's better to access data remotely (from another computer or server) instead of storing it on your own computer.

- **Transactional Data Access**: Means getting data from a remote source (like a server or database) when you need it, instead of keeping a copy on your own machine.

Why?

- Your computer might not have enough RAM (memory) to hold all the data at once.

- Some data might be sensitive or private. Thus, instead of downloading everything, you only get the allowed results from the server, which helps protect privacy.

- If you download a copy, it might become outdated. Accessing the server ensures you're using the latest version.

- R is not great for storing or managing large databases. So, it’s better to let a proper database server handle the data and let R just request what it needs.

JSON

- JSON (**JavaScript Object Notation**) is a simple format of organizing and sharing data so computers can **send it to each other**.

- **Why apps use JSON:**
  - It is **easy to read**
  - It is **easy to send over the internet**
  - Different apps understand it the same way

JSON Data Access

```r
library(RCurl)
library(jsonlite)
bikeURL <-
     'https://gbfs.citibikenyc.com/gbfs/en/station_status.json'

apiResult <- getURL(bikeURL)
results <- fromJSON(apiResult)
str(results)
```

- `RCurl` is a **R package** that helps you **connect to websites** and download data from the internet.
- **jsonlite** is an **R package** that helps you work with **JSON data**. jsonlite lets R **read JSON from the web** and **turn it into R objects** you can work with.
- It also lets you **write R data into JSON** if you want to send it somewhere.

`bikeURL <- 'https://gbfs.citibikenyc.com/gbfs/en/station_status.json'`

- This is the **web address of a JSON file**
- It contains **real-time information about Citi Bike stations in NYC** (like which stations have bikes available, which are empty, etc.)

apiResult <- getURL(bikeURL)

- **getURL()** is from the RCurl package
- It connects to the URL and downloads the content
- The data from the URL is stored in the variable apiResult as plain text (still in JSON format, not yet a table)

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results <- fromJSON(apiResult)

- **apiResult** is JSON text that came from the internet
- **fromJSON()** reads that JSON, converts it into usable R data (lists and data frames), and stores it in results

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The End